

# Aradhya Khandelwal

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[github.com/arukh281](https://github.com/arukh281) | [LinkedIn](#) | [Google Scholar](#) | [aradhya-khandelwal.vercel.app](https://aradhya-khandelwal.vercel.app) | [YC Paxel builder report](#)

ML engineer for multichannel audio: a drone detector I trained only on synthetic data caught 89% of drone segments in 40 real field flights. Co-founder and CTO of an AI startup; two papers and a patent application.

## Skills

**Audio & Signal Processing:** Microphone-array processing, beamforming (SRP-PHAT), GCC-PHAT / TDOA, spectral whitening, harmonic (HPS) analysis, synthetic data generation, sim-to-real transfer

**Machine Learning:** PyTorch, CNNs (1D/2D/3D), ablation studies, scikit-learn, librosa, LLM agents (OpenRouter)

**Systems:** Python, CUDA, NVIDIA Jetson Orin, FastAPI, PostgreSQL, Docker, AWS (EC2, S3), GitHub Actions, Linux

**Web:** TypeScript, React, Next.js

## Experience

**Deep Learning Trainee, DeepTech Division — SSS Defence, Bangalore** Mar 2026 – Present

- Built a deep-learning acoustic drone detector for a multi-microphone array, trained only on synthetic data. On 40 real field flights it caught 89.4% of drone segments (95% CI 87.7–90.9%), even with 15–30 dead microphones; synthetic AUROC 0.995.
- Built the synthetic data generation pipeline: multirotor acoustic synthesis, per-microphone array realism (gain, jitter, dead channels), and time-aligned labels.
- Diagnosed the model with branch knock-out ablations, and added synthetic machinery hard negatives that cut look-alike scores from 0.79 to 0.04–0.16 on synthetic tests.
- Built and debugged a CUDA signal-processing front-end (whitening, cross-spectra, GCC-PHAT, beamforming), ported the detector to NVIDIA Jetson Orin with runtime tests and benchmarks, and built a Raspberry Pi field-recording rig.

**Co-founder & CTO, The Algorithm Lab (early-stage, not yet incorporated)** Jun 2026 – Present

- Only engineer at an AI marketing startup that runs a managed marketing service. Building the product that automates that service, now in beta.
- Built 7 platform-writer agents, a topic planner, an adversarial critic and a 4-stage campaign planner, all on one company audit.
- Built the website-audit pipeline: HTML extraction, headless Playwright screenshots, Claude vision grading and PDF reports.
- Designed a provider layer on OpenRouter, with Claude as the primary model and Gemini as a per-workload fallback. FastAPI, async SQLAlchemy with PostgreSQL and React 19 on Render; 594 automated tests.

**Automation Intern, Intact Group, Gurgaon** Jul 2025 – Mar 2026

- Built Playwright/Python automation for 25+ regression workflows, cutting manual QA time by ~40%, and wired automated validation into CI/CD during the consolidation of 3 subsidiary firms.

**Research Intern, IIT Roorkee** Jun 2025 – Jul 2025

- Built 1D/2D/3D CNNs for 3-class PolSAR and multispectral classification on 11-band satellite data, with band alignment, normalisation, ROI extraction and band-level ablations. Co-authored the IEEE InGARSS 2025 paper below.

## Projects

**Earmark — in-browser personal voice isolation for voice agents** | [\[GitHub\]](#) *in progress* Sep 2026 – Present

- Causal 10 ms-hop model that keeps only the enrolled voice and gates barge-ins, on a C++17 engine compiled to WebAssembly.

**Hypercharge — toolkit for AI coding agents** | [\[GitHub\]](#) Jun 2026 – Jul 2026

- Open-source Python CLI that wires a repo for Cursor and Claude Code agents: a tree-sitter code graph they query before editing, session memory with SQLite full-text search, and advisory safety hooks; 260 automated tests, MIT licence.

**Sensor Anomaly — real-time anomaly detection** | [\[GitHub\]](#) Feb 2026

- Isolation Forest anomaly detection on NASA C-MAPSS turbofan sensor streams, served by a FastAPI streaming API.

**KALI — NSE paper-trading sandbox** | [\[GitHub\]](#) 2026

- NSE paper trading: 4 isolated algorithm ledgers, a scheduled post-close job, backtests from 2018, HMM regimes, Kelly sizing.

**Review Insights — LLM review analytics** | [\[GitHub\]](#) Jan 2026 – May 2026

- LLM pipeline that sorts Google Business Profile reviews into 6 subject buckets with health scores; tested on 150 real reviews.

**SwiftShare — anonymous encrypted file sharing** | [\[GitHub\]](#) Feb 2025 – May 2025

- FastAPI and S3 service with AES-256-GCM, optional passwords and expiring links; basis of my ICDSA paper and patent.

## Publications & Patent

**Unsupervised Approach for Classifying Volume Scattering for Surface Roughness and Potential Water-Ice Deposits**

S. Muduli, A. Khandelwal, A. K. Maurya, S. K. Singh, D. Singh. *IEEE InGARSS 2025*, pp. 594–598. [\[IEEE Xplore\]](#)

**Beyond Nonce Uniqueness: A Cryptographic Evaluation of AES-GCM in Secure Anonymous File Sharing Systems**

A. Khandelwal, L. Upadhyay, S. Shringi. *ICDSA 2025*, Springer, pp. 323–334. [\[Springer\]](#)

**A Secure Anonymous File Sharing System with Optional Password-Based Encryption and Expiry Controls**

Indian Patent Application No. 202511066102 A, published 25 Jul 2025. Co-inventor with Dr. S. Shringi, Manipal University Jaipur.

## Education

**Manipal University Jaipur — B.Tech, Computer Science** Sep 2022 – May 2026

CGPA 8.16/10 | Coursework: Deep Learning, Digital Signal Processing

**The Espee Global School, Roorkee — Class XII (PCM) 95%, Class X 98%** Apr 2019 – Mar 2022

## Leadership

**Convener, Startup Weekend Jaipur (Google for Startups)** Apr 2024 – Nov 2024

Led a 150-member organising team for a 54-hour hackathon with 160+ participants.